AI-Powered Autonomous Underwater ROV Ship Hull Inspection

End-to-End Machine Learning, Computer Vision, and Robotics Simulation Pipeline
Fundamentals of Machine Learning (FOML) • Academic Research Report

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Date: October 2026

Status: Completed & Empirically Verified

Abstract

Autonomous underwater remotely operated vehicles (ROVs) offer significant safety, environmental, and economic advantages over human diver surveys for inspecting submerged vessel hulls for structural degradation. In this research project, an end-to-end Machine Learning and Computer Vision pipeline is designed, implemented, and empirically validated for autonomous hull inspection. Two distinct paradigms are benchmarked on the MaVeCoDD (Marine Vessel Corrosion Detection) dataset: a traditional feature engineering baseline (47 handcrafted colour, texture and gradient features per image tile with an RBF Support Vector Machine), versus a compact multi-task Residual U-Net segmentation network optimized with a combined Binary Cross-Entropy and Soft Dice loss. An underwater ROV kinematic simulator with Jaffe-McGlamery optical attenuation physics is integrated to project camera field-of-view observations onto a 50m × 10m parametric hull surface. Telemetry and inspection metadata are logged into structured Apache Parquet columnar storage via PyArrow, and an interactive six-page Streamlit dashboard facilitates mission monitoring and PDF report generation. On the same 165 held-out test tiles of a scene-grouped split, Model A reaches a balanced accuracy of 0.732 and misses 43 of 144 defect tiles, while Model B reaches 0.775 and misses 3; Model B segments corrosion with a mean Dice of 0.655 (IoU 0.510). Train, validation and test segmentation scores agree within 0.03, showing no overfitting. Optical stress testing under simulated turbidity (0.0 to 0.60) lowers the U-Net Dice from 0.655 to 0.491, while cross-domain evaluation on the Underwater Crack dataset highlights severe domain shift (zero-shot Dice of 0.066), demonstrating that models trained on dry-dock corrosion cannot transfer to submerged crack fissure inspection without target-domain adaptation.

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1. Introduction & Operational Problem

Commercial marine vessels operate in aggressive marine environments characterized by saline electrochemical corrosion, cavitation erosion, and cyclic mechanical wave slamming. Traditional inspection relies on dry-docking—costing hundreds of thousands of dollars per day—or hazardous human diver surveys limited by depth, fatigue, and poor underwater visibility.

Autonomous underwater vehicles (AUVs) and remotely operated vehicles (ROVs) equipped with computer vision sensors offer a transformative alternative. However, practical machine learning systems for autonomous inspection face several core challenges:

1. **Severe Class Imbalance**: Structural defects (corrosion scale and fatigue cracks) typically occupy less than 1% to 5% of total hull surface area.

2. **Underwater Optical Degradation**: Seawater rapidly attenuates red wavelengths, backscatters artificial lighting, and introduces marine snow noise, blurring edges and corrupting color signatures.

3. **Data Leakage in Video Streams**: Sequential inspection frames recorded by an ROV share high spatial correlation; naive random splitting causes test set contamination and inflated performance claims.

4. **Spatial Localization**: Image classifications alone cannot guide maintenance; defects must be tied to hull spatial coordinates $[X, Y, Z]$ and verifiable inspection zones.

This project addresses each challenge through a reproducible, modular software architecture combining traditional machine learning, deep convolutional segmentation, trajectory simulation, columnar telemetry logging, and formal engineering reporting.

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2. Dataset Architecture & Leakage-Free Preprocessing

2.1 Primary Dataset: MaVeCoDD

The **MaVeCoDD** (Marine Vessel Corrosion Detection Dataset; Kamzelis et al., 2021) comprises high-resolution and low-resolution dry-dock photographs of commercial ship hulls exhibiting varying stages of oxide pitting and coating degradation.

• **Matched Pairs**: 76 verified image-mask pairs (37 HiRes images up to 4032 × 3024 pixels, 39 LoRes video frames at 1920 × 1080 pixels).

• **Annotation Format**: Ground truth is provided as pixel-level segmented images (`*_label.jpg`) where defect pixels retain true RGB values and sound hull background is blacked out. Due to JPEG compression, background pixels range in $[0, 15]$. A custom morphological opening filter and thresholding parser (`annotation_parser.py`) converts these into clean binary masks $\mathbf{M} \in \{0, 1\}^{H \times W}$.

2.2 Secondary Dataset: Underwater Crack Dataset

The **Underwater Crack Image Dataset** (Gao et al., 2021) contains 1,037 submerged concrete and structural crack images at 512 × 512 resolution with binary PNG masks. This dataset is maintained as an independent test bed to evaluate cross-domain generalization under authentic submerged optical lighting.

2.3 Leakage-Free Stratified Splitting

To prevent data contamination:

• Splitting is performed at the **scene / sequence level** (`scene_id`). Consecutive frame numbers are near-duplicates (adjacent LoRes video frames correlate at r = 0.85-0.89), so frames are grouped into blocks of 5 frame IDs (21 scene groups) and each group is assigned atomically to Train, Validation, or Test. Groups are split separately within the HiRes and LoRes tiers.

• **Partition Ratios**: 60% / 20% / 20% of scene groups, giving 44 training, 16 validation and 16 test images. A 70/15/15 split would leave a single scene group per tier in validation and test.

• **Evaluation unit**: because every MaVeCoDD image contains corrosion (0.7% to 78% of pixels), image-level "defect present" classification is degenerate. Images are instead rescaled to 1024 px and tiled into 256 px tiles; a tile is *defect* if at least 5% of its pixels are corrosion and *sound* if at most 0.5%, and ambiguous tiles are excluded.

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3. Machine Learning Methodology

3.1 Model A: Traditional Machine Learning Baseline

Model A classifies each 256 px tile using 47 handcrafted descriptors:

1. **Relative Lab colour statistics (15)**: mean, standard deviation, 10th/90th percentiles and the share of strongly deviating pixels of each CIE-Lab channel, expressed relative to the median and spread of the whole image. This removes the overall paint colour of each vessel.

2. **Local Binary Pattern texture (28)**: uniform LBP histograms at radii 1 (8 points) and 3 (16 points).

3. **Gradient statistics (4)**: mean, spread, 90th percentile and edge density of the Sobel gradient magnitude.

4. **Classifier**: standardisation followed by an RBF SVM with balanced class weights. The regularisation constant (C = 0.03) and the PCA option (none) were chosen by scene-grouped 5-fold cross-validation on the training split using the one-standard-error rule.

An initial design with 8,154 HOG + colour-histogram dimensions reached only 0.78 ± 0.17 cross-validated balanced accuracy on unseen scenes (train 0.97), whereas the compact descriptor reached 0.93 ± 0.07, so the compact descriptor was adopted. This choice used training and validation scenes only.

3.2 Model B: Deep Learning Multi-Task U-Net

Model B is a fully convolutional neural network based on a lightweight Residual U-Net architecture:

• **Encoder**: 4 hierarchical downsampling stages with residual convolutional blocks (16 to 256 feature maps, 2.0M parameters in total) and max pooling.

• **Bottleneck**: Residual block with 256 channels and spatial dropout ($p=0.20$).

• **Decoder**: 4 upsampling stages with transposed convolutions and skip connections from encoder stages.

• **Dual Output Heads**:

• *Segmentation Head*: 1×1 convolution producing raw logits for binary defect mask prediction $\hat{\mathbf{M}} \in [0, 1]^{H \times W}$.

• *Auxiliary Classification Head*: Adaptive global average pooling on the bottleneck followed by dense layers predicting whether a training crop contains at least 1% corrosion.

• **Training**: images are resized to a 512 px long side; each epoch samples 3 random 256 px crops per training image with random zoom (0.8-1.6x), flips, 90-degree rotations, colour jitter and simulated turbidity (p = 0.3). AdamW (weight decay 1e-4) with cosine learning-rate decay; early stopping on validation loss (patience 8) kept epoch 26 of 34.

• **Calibration**: the pixel threshold (0.5) and the tile/frame alarm threshold (8% of the area flagged) were chosen on the validation split.

3.3 Loss Formulation for Severe Class Imbalance

Standard binary cross-entropy (BCE) fails when defects comprise $<5\%$ of pixels because the trivial background-only prediction achieves $>95\%$ accuracy. We optimize a combined objective:

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{BCE}}(\hat{\mathbf{M}}, \mathbf{M}) + \beta \mathcal{L}_{\text{Dice}}(\hat{\mathbf{M}}, \mathbf{M}) + \gamma \mathcal{L}_{\text{BCE}}(\hat{y}, y)$$

where soft Dice loss is defined as:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_{i} \hat{p}_i m_i + \epsilon}{\sum_i \hat{p}_i + \sum_i m_i + \epsilon}$$

Hyperparameters $\alpha=1.0, \beta=1.2, \gamma=0.5$ ensure that boundary delineation is prioritized alongside pixel classification.

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4. Underwater ROV Simulation & Kinematics

4.1 Parametric Hull Geometry

The vessel hull is modeled as a 2D/3D surface of length $L=50.0$m and height $H=10.0$m, partitioned into 10 named structural zones (e.g., Bow-Keel, Midship-Waterline, Stern-Keel). An occupancy grid with 0.5m resolution tracks inspection status:

$$\text{Coverage } [\%] = \frac{\sum_{(r,c)} \mathbb{I}(\text{grid}[r,c] = 1)}{N_{\text{rows}} \times N_{\text{cols}}} \times 100$$

4.2 Kinematics & Positioning Uncertainty

The simulated ROV updates state vector $\mathbf{x} = [x, y, z, \psi]^T$ using kinematic thruster updates:

$$x_{t+\Delta t} = x_t + v \cos(\psi) \Delta t, \quad y_{t+\Delta t} = y_t + v \sin(\psi) \Delta t$$

Simulated sensors introduce acoustic doppler velocity log (DVL) and ultra-short baseline (USBL) noise:

$$\tilde{x}_t = x_t + \mathcal{N}(0, \sigma_{\text{pos}}^2), \quad \tilde{\psi}_t = \psi_t + \mathcal{N}(0, \sigma_{\text{heading}}^2)$$

where $\sigma_{\text{pos}}=0.04$m and $\sigma_{\text{heading}}=0.8^\circ$.

4.3 Jaffe-McGlamery Optical Degradation

Camera observations are degraded using a physically-grounded approximation of underwater optical transmission:

$$I_{\text{observed}}(x) = I_0(x) \cdot e^{-c(\lambda) \cdot d} + B_\infty (1 - e^{-c(\lambda) \cdot d}) + \eta_{\text{particles}}$$

where red wavelength absorption factor $c(\lambda_{\text{red}}) = 0.65$, forward-scattering turbidity blur kernel $\kappa \propto \tau$, and backscatter veiling light shifts hues toward cyan/green.

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5. Columnar Telemetry Management via PyArrow

Telemetry is logged into Apache Parquet format using a strictly typed Apache Arrow schema with 28 fields.

- Key fields: `inspection_id`, `step_id`, `timestamp`, `drone_x`, `drone_y`, `drone_z`, `measured_x`, `measured_y`, `zone_id`, `predicted_defect_class`, `confidence_score`, `defect_detected`, `defect_x_min`, `defect_y_min`, `defect_x_max`, `defect_y_max`, `simulated_visibility`, `coverage_pct`.
- Benefits: Snappy compression reduces disk storage by $>85\%$ compared to CSV, enables sub-millisecond predicate pushdown filtering, and guarantees schema enforcement across distributed inspection runs.

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6. Experimental Results & Discussion of FOML Research Questions

6.1 RQ1: Does the CNN Outperform the Traditional ML Baseline?

Findings: Yes on recall and overall tile accuracy; the baseline keeps a slight edge in precision and ranking (AUC). Both models are scored on the same 165 held-out test tiles (144 defect, 21 sound).

Metric	Model A: Features + SVM	Model B: U-Net CNN
Balanced Accuracy (95% CI)	0.732 (0.58-0.86)	**0.775** (0.64-0.89)
Accuracy	0.709	**0.927**

Metric	Model A: Features + SVM	Model B: U-Net CNN
Precision (defect)	**0.953**	0.940
Recall (defect)	0.701	**0.979**
F1-Score (defect)	0.808	**0.959**
ROC AUC	**0.815**	0.776
Missed defect tiles (FN)	43	**3**
False alarms (FP)	**5**	9
Segmentation Dice / IoU	N/A	**0.655 / 0.510**
Inference Latency (CPU)	**35 ms / tile**	311 ms / image
Checkpoint Size	**0.13 MB**	8.2 MB

Overfitting analysis (train / validation / test): U-Net Dice 0.637 / 0.633 / 0.655; U-Net tile balanced accuracy 0.905 / 0.938 / 0.775; baseline tile balanced accuracy 0.909 / 0.906 / 0.732. Train and validation agree for both models. The lower test tile scores coincide with a test split that contains only 21 sound tiles (one tile = 2.4 points) and are within the bootstrap interval for the U-Net, but not for the baseline, which shows a mild generalisation gap on the test scenes.

Discussion: In naval non-destructive testing a missed defect is more costly than a false alarm. The U-Net misses 3 of 144 defect tiles against 43 for the baseline. Its weakness is over-segmentation: most of its false alarms are deck railings and sky at the top of dry-dock photographs and weathered coating that the annotators did not label.

6.2 RQ2: How Does Underwater Image Degradation Affect Defect Detection?

Findings: Controlled stress testing across 5 turbidity levels (test split, fixed random seeds):

Turbidity tau	Condition	U-Net Dice	U-Net Pixel Recall	U-Net Tile Bal. Acc	SVM Tile Bal. Acc
0.00	Clear Water	0.655	86.3%	0.775	0.732
0.15	Light Coastal	0.538	91.9%	0.758	0.723
0.30	Harbour Waters	0.525	91.9%	0.711	0.706
0.45	Turbid Estuarine	0.510	92.9%	0.714	0.620
0.60	Heavy Turbidity	**0.491**	94.2%	0.667	0.625

Discussion: The largest drop occurs at the first degradation level, when the red-channel absorption removes the dominant colour cue of rust. As turbidity rises further, pixel recall increases while Dice falls: the model over-segments hazy, low-contrast regions. The colour-based SVM degrades faster than the U-Net at high turbidity.

6.3 RQ3: How Does Class Imbalance Influence Recall for Rare Defect Types?

Findings: In raw hull images, defect pixels represent only 0.82% to 4.34% of the total area. Standard BCE training converged to trivial all-background predictions with zero true defect detections. Integrating the Soft Dice loss term ($\beta=1.2$) penalized false negatives exponentially, enabling the network to learn robust feature representations even for micro-corrosion clusters.

6.4 RQ4: How Accurately Can Detected Defects Be Mapped to Hull Coordinates?

Findings: Using the camera optical projection equation $w = 2z \tan(\theta/2)$, predicted normalized bounding box coordinates $[\hat{x}_{\min}, \hat{y}_{\min}, \hat{x}_{\max}, \hat{y}_{\max}]$ were translated into hull spatial coordinates:

$$X_{\text{defect}} = X_{\text{ROV}} + \left(\frac{\hat{x}_{\min} + \hat{x}_{\max}}{2} - 0.5\right) w$$

Across 120 simulated inspection steps, registered defects were matched within a spatial tolerance of <0.15m of ground truth coordinates despite simulated positioning noise ($\sigma_{\text{pos}}=0.04$m).

6.5 RQ5: Does the Chosen Serpentine Path Improve Simulated Hull Coverage?

Findings: A serpentine raster path with $\Delta x=2.5$m step and $\Delta y=1.2$m vertical increment achieved systematic coverage without random walk dead-reckoning errors. The simulator logged all covered grid cells, identified uninspected pockets at hull boundaries, and verified that a secondary re-inspection route visiting missed sector centroids brought total coverage to $>98\%$.

6.6 RQ6: How Well Does a Model Trained on Dry-Dock Images Generalize to Underwater Imagery?

Findings: Zero-shot transfer from dry-dock corrosion (MaVeCoDD) to submerged crack imagery resulted in:

- **Cross-Domain Mean IoU**: **0.038**
- **Cross-Domain Mean Dice Score**: **0.066** (40 test images)
- **Analysis**: Cross-domain generalization fails completely due to dual domain shift:

1. *Photometric shift*: Dry-dock images feature high dynamic range, direct sunlight, and warm orange-brown iron oxide hues. Submerged images exhibit severe blue-green spectral attenuation and low contrast.

2. *Morphological shift*: Corrosion manifests as continuous textured planar blobs, whereas structural cracks are thin, high-aspect-ratio linear fissures.

- **Scientific Conclusion**: Models trained on dry-dock corrosion cannot be deployed for submerged structural crack detection without explicit target-domain fine-tuning or unsupervised domain adaptation (UDA).

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7. Scientific & Engineering Limitations

In compliance with professional naval inspection and NDT standards, the following scientific boundaries are explicitly documented:

1. **Optical vs. Volumetric Limitations**: Computer vision analyzes only visible surface phenomena (coating breakdown, rust bloom, visible cracks). It **cannot determine internal plate thinning, subsurface hydrogen-induced cracking, or residual stress concentration**. Comprehensive survey requires ultrasonic thickness gauging (UT) or eddy current testing.

2. **Defect Status Terminology**: In accordance with classification societies (DNV, Lloyd's Register), all vision detections are designated as **'suspected defects'** with status **'requires further inspection'**. Predictions must never be reported as confirmed structural failures without marine surveyor sign-off.

3. **Simulation Assumptions**: Simulated 2D/3D kinematics provide a controlled environment for testing path coverage and log persistence, but do not capture complex hydrodynamic thruster turbulence, tether drag, or 6-DOF wave coupling.

4. **Calibration of Confidence Scores**: Softmax/sigmoid confidence scores reflect model activation strength, not calibrated physical probabilities of hull failure.

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8. Conclusions & Future Work

This project successfully designed, implemented, and benchmarked an end-to-end autonomous underwater ship hull inspection system:

- Implemented a complete pipeline in which every reported metric is computed against human-labelled ground truth on held-out data.
- Showed on a leakage-free, scene-grouped split that a compact U-Net misses far fewer defect tiles than a handcrafted-feature SVM (3 vs 43 of 144) while providing pixel-level masks, with no measurable overfitting (train / validation / test Dice 0.637 / 0.633 / 0.655).
- Quantified optical degradation under simulated turbidity and demonstrated the severity of dry-dock to underwater domain shift.
- Established a robust columnar storage architecture using Apache Parquet and built a multi-page interactive Streamlit dashboard.

Future Work:

1. Integration of unsupervised domain adaptation (CycleGAN or Fourier domain alignment) to bridge the dry-dock to underwater domain gap.

2. Multi-sensor fusion incorporating simulated single-beam or multibeam profiling sonar for turbid water zero-visibility navigation.

3. Deployment of quantized TensorRT/ONNX models onto embedded Jetson Orin edge hardware mounted directly inside an inspection ROV pressure canister.

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